

Brazil Exit Node Family Controller

AWS São Paulo + Tailscale + Flutter

A practical step-by-step companion to the original two-page Tailscale AWS Exit Node Manual, adapted for an existing tailnet and two family phones.

Scope: Amazon Linux 2023 EC2 in sa-east-1, Cognito sign-in, API Gateway/Lambda power control, Flutter starter, local desktop routing automation and explicit mobile limitations.

Critical distinction: The round button changes EC2 power state. It does not globally set Tailscale routing. A pinned mobile client can lose Internet if the selected exit node is stopped; this guide keeps that behavior visible rather than hiding it.

Companion package: CloudFormation deployment template, Flutter source, and optional desktop route agent. Nothing is deployed automatically by this document.

1 Summary and feasibility

You already have Tailscale configured on some computers and on an Android phone, so this guide starts with the missing infrastructure. The Brazilian EC2 machine becomes one additional member of your existing tailnet.

Desired behavior	Feasibility	Implementation
Start/stop one AWS instance from both phones	YES	Flutter app + Cognito + API Gateway + Lambda
Automatically select Brazil exit on desktop	YES, with extra local software	A per-computer helper polls Tailscale state and updates local routing
Automatically revert desktops to local Internet	BEST-EFFORT	Agent clears selected exit node when it becomes unavailable
Change native iPhone/Android Tailscale setting using Flutter button	NOT THROUGH A DOCUMENTED TAILSCALE API	Manual Tailscale action; advanced native VPN or device management is a different project
One global Tailnet default exit route	NO	Tailscale exit nodes are selected per client

What counts as ready?

- EC2 running: AWS reports that the virtual machine is powered on.
- Tailscale online: `tailscale status` sees the cloud machine connected to the tailnet.
- Exit approved: Tailscale admin allows that machine to serve as an exit node.
- Routing active: this particular client has selected the exit node and an external IP check shows AWS Brazil.

Recommended rollout: First provision AWS and manually validate one laptop. Second deploy the secure phone button. Third introduce desktop automation. Keep phones manual initially; then evaluate whether you actually need an advanced router or VPN integration.

2 Architecture and traffic paths

Control plane: the power button communicates with AWS even while EC2 is stopped. The AWS API must not run on the EC2 instance itself.

ANDROID / IPHONE	COGNITO IDENTITY	API GATEWAY + LAMBDA	EC2 POWER API
Flutter sign-in + button	Two invited users only	JWT-authenticated HTTPS	Start / Stop / Describe

Data plane when Brazil is selected: a device tunnels ordinary outbound Internet traffic to its Tailscale connection on the Brazil host; that host NATs the traffic to the public Internet. The destination normally sees the EC2 public IP.

```
Your device --> Tailscale tunnel --> EC2 (sa-east-1) --> Internet
+----- Tailnet coordination -----+
```

Data plane when no exit node is selected: the device sends Internet traffic through its usual Wi-Fi router, Ethernet gateway, or cellular provider. Tailscale peer-to-peer access can remain enabled.

```
Your device --> Local Wi-Fi router / cellular --> Internet
Your device --> Tailscale peers (private traffic still possible)
```

Routing policy and travel

- The Brazil exit node is useful for appearing to originate from Brazil. It adds round-trip latency, can reduce throughput, and does not guarantee every streaming service will accept a data-center IP.
- Allow LAN access in the Tailscale exit settings if you need the device to reach local printers or LAN equipment; this does not replace its Internet exit selection.
- Your spouse can control EC2 with the Flutter app, but must also be able to access the existing tailnet and choose the exit node separately in Tailscale.

3 AWS prerequisites

- AWS account with billing alerts and an operator identity authorized to create CloudFormation, EC2, IAM, Lambda, API Gateway, Cognito and VPC security groups.
- Region: South America (São Paulo), sa-east-1. Always check the region selector before creating resources.
- An existing VPC and public subnet. Its route table must direct 0.0.0.0/0 to an Internet Gateway (IGW). An ordinary private subnet without NAT/egress will not work here.
- AWS CLI v2 installed on an administrator machine. Flutter SDK needed later, plus Xcode on a Mac to deploy to your spouse's iPhone.

Identify a default VPC and subnet

```
aws ec2 describe-vpcs --region sa-east-1 \
  --filters Name=isDefault,Values=true \
  --query "Vpcs[*].[VpcId,CidrBlock]" --output table

aws ec2 describe-subnets --region sa-east-1 \
  --query "Subnets[*].[SubnetId,VpcId,AvailabilityZone]" \
  --output table
```

If the AWS account has no default VPC, create or select a suitable VPC and Internet Gateway first. Do not deploy into a subnet until its route table is checked. A public IP alone is not sufficient.

Security posture

- No open inbound SSH port; administration uses AWS Systems Manager Session Manager. Amazon Linux 2023 generally has the SSM Agent preinstalled.
- EC2 has only the AmazonSSMManagedInstanceCore role, not credentials to start/stop other instances.
- The Lambda role can start/stop exactly one EC2 instance. The app receives no AWS keys and uses short-lived Cognito access tokens.
- Keep AWS account root login, cloud credentials, Tailscale auth keys and mobile app binaries private. Consider MFA on both AWS and Cognito.

4 Provision the complete AWS stack

The companion file `aws/stack.yaml` declares EC2, an all-egress/no-ingress security group, Systems Manager instance profile, Cognito user pool and classic Hosted UI, authenticated HTTP API, and one Lambda controller.

```
aws cloudformation deploy \
  --region sa-east-1 \
  --stack-name brazil-tailnet \
  --template-file aws/stack.yaml \
  --capabilities CAPABILITY_IAM \
  --parameter-overrides \
    VpcId=vpc-REPLACE \
    PublicSubnetId=subnet-REPLACE \
    DomainPrefix=replace-with-unique-lowercase-prefix
```

The Cognito domain prefix must be globally unique in its namespace. The stack can take several minutes to create. The EC2 node boots and installs Tailscale but is not automatically enrolled in your tailnet, so no auth key is exposed through CloudFormation outputs.

Record the outputs

```
aws cloudformation describe-stacks \
  --region sa-east-1 --stack-name brazil-tailnet \
  --query "Stacks[0].Outputs[*].[OutputKey,OutputValue]" \
  --output table
```

Save `Ec2InstanceId`, `UserPoolId`, `AppClientId`, `CognitoIssuer` and `ApiBaseUrl`. The latter three configure the Flutter app, while the instance ID is used for troubleshooting.

Server bootstrap assumptions

- Amazon Linux 2023 x86_64; Tailscale installed from its official install script.
- IPv4 and IPv6 kernel IP forwarding enabled persistently in `/etc/sysctl.d/99-tailscale-exit.conf`.
- Dynamic public IPv4 only, allocated at instance start. No Elastic IP; no open inbound ports.
- Outbound HTTPS access is required for the control plane and SSM; outbound UDP can improve direct Tailscale connectivity. Relay fallback may be slower.

If the stack fails, open CloudFormation > Events. Common blockers are a domain prefix already in use, an incompatible subnet/VPC pairing, IAM restrictions, or an AWS quota. Do not blindly repeat deployments with new stacks.

5 Join EC2 to your existing tailnet

In AWS Console, open EC2 > Instances > brazil-tailscale-exit > Connect > Session Manager. Alternatively, with correct SSM permissions and a local Session Manager plugin:

```
aws ssm start-session --region sa-east-1 \  
  --target i-REPLACE_WITH_STACK_OUTPUT
```

Once inside the instance:

```
sudo systemctl status tailscaled --no-pager  
sudo sysctl net.ipv4.ip_forward  
sudo sysctl net.ipv6.conf.all.forwarding  
sudo tailscale up --advertise-exit-node  
sudo tailscale status
```

The first `tailscale up` command should display an authentication URL. Open it in a browser while signed in as the correct tailnet owner, approve the new computer, then return to the terminal.

Approve exit-node advertisement

- In <https://login.tailscale.com/admin/machines>, locate `brazil-tailscale-exit`.
- Open its route settings and enable `Use as exit node`.
- Review your tailnet access grants/ACLs. Exit-node clients require permission to `autogroup:internet`, not merely access to EC2's machine IP.
- Consider disabling device key expiry for this dedicated connector, understanding the security trade-off; key expiry while offline may make routes unavailable and clients fail closed.

First manual functional test

- On just one existing computer, open its Tailscale exit-node setting and select the Brazil EC2 node. Allow LAN access if needed.
- Visit <https://ifconfig.me> or another public-IP tester. Verify the egress IP matches EC2's current public IPv4 and appears geolocated to Brazil (commercial geo databases may lag).
- Before stopping EC2, select `None` in the test client. Verify its original public IP appears again.
- Stop and restart EC2. Confirm Tailscale reconnects under the same tailnet identity and can again serve as exit node.

6 Phone button: authenticated controller

This is deliberately a minimal app: first-run authentication, then a prominent round toggle and small AWS state label. EC2 operations are asynchronous; tapping ON is a request, not a guarantee that Tailscale is ready.

Layer	Responsibility
Flutter UI	Single power button; read EC2 state; show transitions and errors
Cognito user pool	Two invited family members sign in separately; authorization-code / PKCE
API Gateway HTTP API	Validate JWT issuer, audience and openid scope
Lambda	GET /state; POST /power; call EC2 start/stop/describe
EC2	Actual Brazilian VPN egress when Tailscale is running

Why not put AWS keys in Flutter?

An Android APK or iOS binary can be reverse engineered. Long-lived AWS access keys embedded in an app would expose privileged instance-control credentials. This design uses identity tokens and an AWS-managed least-privilege backend instead.

One-time identity provisioning

After deployment, create one Cognito user for yourself and one for your wife. The stack disables public self-registration:

```
aws cognito-idp admin-create-user \  
  --region sa-east-1 \  
  --user-pool-id YOUR_POOL_ID \  
  --username FIRST_EMAIL \  
  --user-attributes Name=email,Value=FIRST_EMAIL \  
                    Name=email_verified,Value=true \  
  --desired-delivery-mediums EMAIL
```

Repeat for the second email. Each recipient follows Cognito's sign-in/password-setup process. Check user delivery configuration and quotas if invitation messages do not arrive.

Controller endpoint contract

```
GET /state -> {"state":"running", "exitNodeReady":null}  
POST /power {"action":"on"} -> request start  
POST /power {"action":"off"}-> request stop
```

The value `exitNodeReady:null` is intentional. Checking EC2 state does not establish whether the Tailscale agent has started or whether an individual device is routing through it.

7 Build Flutter for Android and iPhone

The bundle includes `flutter/lib/main.dart`, `flutter/pubspec.yaml` and a detailed setup file. Its sources use `flutter_appauth`, `flutter_secure_storage` and `http`. No compiled APK or IPA is claimed.

Generate a standard Flutter project

```
flutter create --org dev.family brazil_exit_button
cd brazil_exit_button
# Copy bundle/flutter/lib/main.dart into lib/main.dart
# Replace pubspec.yaml with bundle/flutter/pubspec.yaml
flutter pub get
```

Register the OAuth callback in both platforms

Cognito's registered callback is `brazilexit://oauth`. Android must set the `appAuthRedirectScheme` manifest placeholder to `brazilexit`, and iOS must register the same URL scheme under `CFBundleURLTypes` in `Runner/Info.plist`. Detailed snippets are in the Flutter README.

Build parameters

```
flutter run \
  --dart-define=API_BASE_URL=https://API_ID.execute-api.sa-east-1.amazonaws.com \
  --dart-define=COGNITO_ISSUER=https://cognito-idp.sa-east-1.amazonaws.com/P00L_ID \
  --dart-define=COGNITO_CLIENT_ID=APP_CLIENT_ID
```

- Android: install the Flutter SDK/Android tooling and use `flutter build apk --release` with the same definitions. Distribute the APK only to trusted devices; keep signing keys under your control.
- iPhone: requires a Mac with Xcode and Apple signing/provisioning. Use Xcode device deployment or TestFlight. You cannot create a correctly signed iOS app from this Linux environment.
- Configuration caveat: public issuer/client IDs and endpoint URL are not secrets, but a correctly signed-in Cognito account is mandatory before any control operation.

Do not test the button with every device pinned to Brazil. If the EC2 host stops and your phone is still using that exit node, the phone may not be able to reach the API to turn it on again until you select None in the Tailscale app.

8 Routing on computers: optional automation

The package includes `desktop/route_agent.py`, a Python proof-of-concept for machines with Tailscale CLI access. It periodically reads local `tailscale status --json` and sets/clears the selected exit based on whether the AWS peer is online and approved for exit use.

Basic setup

- Obtain the Brazil EC2 node's stable Tailscale IPv4 (not its changing AWS public IPv4), for example from the admin console or `tailscale ip -4` on EC2.
- On each desktop, install Python 3 and verify the Tailscale CLI works from that execution environment. Tailscale setting changes can require administrator privileges.

```
AWS_EXIT_IP=100.x.y.z CHECK_SECONDS=20 \  
python3 desktop/route_agent.py
```

- Run interactively for a day before registering as a launchd/systemd task or a Windows Scheduled Task. Prefer a dedicated system account over granting broad privileges to an untrusted script.
- Inspect the generated traffic path and fallback while switching power from another phone. The agent is best-effort, with detection delay, not a perfect failover guarantee.

Why not just use Tailscale auto:any?

Tailscale exposes `--exit-node=auto:any` in its CLI, but auto-selecting a recommended exit node is not equivalent to selecting “Brazil when it exists, ordinary local gateway otherwise.” It can prefer another exit node; if all required exit nodes disappear, mandatory modes can fail closed. Suggested exit node capability also depends on your Tailscale subscription.

Windows, macOS and Linux

Manual selection is available through official Tailscale clients. Automation depends on local OS permissions and CLI semantics. The included agent was statically reviewed but not executed on your machines; treat it as a starting point, not a polished installer.

9 Mobile and whole-LAN alternatives

Native Tailscale on Android / iPhone

Current documented mobile exit-node workflows require a selection in the official Tailscale client. The AWS controller is not itself a VPN provider, and mobile operating systems protect VPN configuration. There is no general remote “change every tailnet client’s exit node” switch.

- Least complex: Flutter turns EC2 on; wait for start; open Tailscale to select Brazil. Before OFF, change Tailscale Exit Node to None. Then tap Flutter OFF.
- Advanced: a custom native VPN app would have to own the tunnel and implement the routing itself. This is a materially bigger and security-sensitive project than a Flutter controller UI.
- MDM / mandatory exit node: supported on some plans, but mandatory exit nodes are unsuitable for simple “OFF means local direct Internet” because offline endpoints can intentionally block traffic.

LAN-gateway approach

A programmable router or dedicated Linux gateway can route a home LAN through a tunnel to São Paulo when EC2 is up and through the ISP when it is down. That can avoid changes on every device while they are on that particular LAN. However, it requires configuring the LAN gateway and robust health checks, and does not solve routing for a mobile phone on cellular or an unrelated Wi-Fi network.

This project currently assumes existing Tailscale clients, not replacement of the household router. Add a gateway phase only if automatic coverage for non-Tailscale televisions/IoT devices matters.

Practical mobile operating order

```
ON : Flutter ON -> wait for EC2/Tailscale -> Tailscale Exit Node = Brazil
OFF: Tailscale Exit Node = None -> verify local IP -> Flutter OFF
```

No silent fallback: A mobile device that remains pinned to an unreachable exit node may lose Internet. The safe sequence is to deselect the exit node before powering down EC2. Desktop automation reduces that risk but does not remove it during failover.

10 Costs, lifecycle and controls

The original manual's part-time cost estimate did not include persistent EBS storage and assumed other expenses were zero. AWS costs vary with region, instance type, storage, usage, feature tiers and account free-tier eligibility; use AWS Pricing Calculator before provisioning.

Item	60 hours/month illustration	Stopped-state behavior
EC2 compute	$60 \times \text{current sa-east-1 on-demand rate}$	Stopped instance compute is not billed
Public IPv4	$60 \times \$0.005 = \0.30	Auto-assigned IPv4 is released; avoid reserved Elastic IP
EBS root volume	$\text{GB-month} \times \text{current EBS storage rate}$	Continues billing while EC2 is stopped
Data transfer	Depends on egress GB and AWS allowances	Traffic charges can apply when used
Cognito / API GW / Lambda / logs	Depends on requests, user pool tier and retention	Some charges can remain independently of EC2

Lifecycle checklist

- Before stopping: disconnect manually managed phones first and let desktop agents reconcile where used.
- After starting: wait for instance running, Tailscale online, exit route available; then test public IP before using sensitive flows.
- Use AWS Budgets or a billing alert. Consider an EC2 auto-stop schedule as a safety limit, but not if family devices are likely to stay pinned overnight.
- No Elastic IP is required: Tailscale node identity survives stop/start on its retained EBS root volume; its AWS public IPv4 can change.
- Deleting the CloudFormation stack terminates the instance and normally deletes the root EBS volume; do not delete without intentionally retiring the Tailscale device.

Indicative example, NOT an AWS quote

At 60 hours/month, the auto-assigned public IPv4 component alone is approximately \$0.30/month. Instance runtime is a separate region-specific hourly charge. An 8 GiB root disk remains billed across the full month. The total can exceed the old manual's "< \$1.20" estimate.

11 Validation and troubleshooting runbook

Symptom	Likely cause / check
EC2 running but no exit option	Tailscale not logged in, not advertising, or route not approved in tailnet admin
Phone button says running, no Brazil IP	App reports only EC2 state; separately confirm Tailscale connection and selected exit
Flutter receives 401 / 403	Missing/expired access token; wrong issuer/client; account user not invited; auth scope mismatch
EC2 cannot access Internet	Public subnet route to IGW, public IPv4, outbound SG, DNS, or OS routing
Session Manager cannot connect	SSM agent, instance profile, outbound reachability, region, operator IAM permissions
Internet stops after EC2 OFF	Selected exit is offline; manually set Exit Node = None, or reconcile via desktop agent
Cognito web login fails	Domain prefix not active, redirect URI mismatch, user invitation/password state
iOS app cannot be installed	Xcode signing, Apple provisioning, bundle ID, or device trust/TestFlight workflow

One-client smoke test

- Provision AWS in São Paulo and enroll EC2 in the original tailnet.
- Approve exit-node usage; test official Tailscale selection on a computer.
- Check external IP shows EC2 public IP; clear exit selection; check ISP IP returns.
- Use AWS CLI/Console to stop/start EC2; verify rejoin and no manual re-approval.
- Invite exactly two Cognito accounts; authenticate app; test ON / OFF while test client is unpinned.
- Only then test per-desktop automation and document fail-close response.

Rollback: If routing is broken, open Tailscale and disable use of the exit node on that device. If needed disconnect Tailscale entirely to regain local Internet temporarily. Do not assume the Flutter app can fix a VPN route on its own.

12 Next improvements and limitations

Useful future upgrades

- Readiness check: add a dedicated health-check channel that can verify Tailscale service and successful Brazilian egress, rather than just checking EC2 state.
- Better power-state orchestration: wait until all enrolled/automated desktops have acknowledged local route removal before stopping. This needs a client management channel; it is not implemented in the starter.
- Hardening: audit logs, AWS rate limiting, alarms, MFA for users, production review of CloudFormation permissions, key rotation and deployment pipeline.
- Optional on/off budget: EventBridge scheduler or idle timer to shut down EC2 after a chosen maximum session, coordinated with routing shutdown.
- Optional LAN-gateway phase: OpenWrt or Linux default gateway with explicit policy-routing and health checks for home-only seamless switching.

What this deliverable includes

- This expanded printable PDF; AWS CloudFormation file; Flutter Dart UI; project-generation and signing guidance; optional Python desktop agent.
- It does not include a deployed AWS stack, provisioned Cognito accounts, tested mobile binaries, an iOS VPN extension, or a working router-level failover service.

Authoritative references

Tailscale exit nodes — <https://tailscale.com/kb/1103/exit-nodes/>

Tailscale CLI and --exit-node options — <https://tailscale.com/docs/reference/tailscale-cli>

Tailscale mandatory exit nodes / fail-close — <https://tailscale.com/docs/features/exit-nodes/mandatory-exit-nodes>

Tailscale suggested exit nodes and plan availability — <https://tailscale.com/docs/features/exit-nodes/auto-exit-nodes>

AWS EC2 stop/start behavior — <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/how-ec2-instance-stop-start-works.html>

AWS EC2 StopInstances billing and retention — https://docs.aws.amazon.com/AWSEC2/latest/APIReference/API_StopInstances.html

AWS VPC public IPv4 pricing — <https://aws.amazon.com/vpc/pricing/>

AWS API Gateway JWT authorizers — <https://docs.aws.amazon.com/apigateway/latest/developerguide/http-api-jwt-authorizer.html>

AWS Cognito CloudFormation app client — <https://docs.aws.amazon.com/AWSCloudFormation/latest/TemplateReference/aws-resource-cognito-userpoolclient.html>

Flutter AppAuth Android and iOS setup — https://pub.dev/packages/flutter_appauth

SSM Agent preinstalled AMIs — <https://docs.aws.amazon.com/systems-manager/latest/userguide/ami-preinstalled-agent.html>

Basis: Original user-provided 2-page Tailscale AWS Exit Node Manual. Additional material reflects public Tailscale, AWS and Flutter documentation consulted in October 2026.